Echo Chambers and Political Efficacy: Algorithmic Curation in Online Discussions

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Abstract: Algorithmic feeds personalize political content, often segregating users into ideologically homogeneous online networks. This paper investigates how algorithmic exposure to partisan echo chambers impacts users' internal political efficacy and willingness to engage in cross-party political dialogue.

1. Research Question

Research Question: Does passive exposure to algorithmically curated partisan echo chambers increase internal political efficacy while decreasing willingness to participate in bipartisan deliberative assemblies?

2. Empirical Methodology & Data Collection

Empirical Methodology & Data Collection: We deployed a custom web browser extension tracking 1,800 active social media users over a 6-month period. Participants were passively monitored for content feed homogeneity (Echo Index: 0–100). Pre- and post-study psychological batteries evaluated Internal Political Efficacy (IPE) and Cross-Party Deliberation Willingness (CPDW) on validated 7-point Likert scales.

Table 1: Empirical Findings Summary

Feed Homogeneity Tier	Sample (n)	Mean Echo Index	Change in IPE (Δ)	Change in CPDW (Δ)
Low Homogeneity (Diverse)	600	22.4	+0.12 (ns)	+0.35 ($p < 0.05$)
Moderate Homogeneity	600	54.8	+0.45 ($p < 0.05$)	-0.28 ($p < 0.05$)
High Homogeneity (Echo Chamber)	600	86.1	+1.24 ($p < 0.001$)	-1.42 ($p < 0.001$)

3. Discussion & Implications

Discussion & Conclusion: Our findings reveal a dual empirical effect of algorithmic curation. High feed homogeneity significantly increases internal political efficacy (+1.24 points), reinforcing users' confidence in their political knowledge. However, this comes at a steep cost to civic deliberation, causing a sharp drop (-1.42 points) in willingness to engage with opposing political views, fostering affective polarization.